

Six effects shaping the detected dispersion-measure distribution of fast radio bursts

A pedagogical toy-model walk-through

Single-dish surveys: ASKAP-like ($D = 12$ m), Parkes-like ($D = 60$ m), FAST-like ($D = 300$ m).

Summary

We build a deliberately minimal model of the cosmological FRB population, adding one effect at a time, to expose how each reshapes the detected dispersion-measure (DM) distribution. The five dominant *astrophysical* effects are (1) **Euclidean geometry**, (2) **the luminosity function**, (3) **cosmology**, (4) **the cosmic star-formation history**, and (6) **the intrinsic spectral index**; we also include (5) **intra-channel DM smearing**, the dominant *instrumental* effect on FRB sensitivity at high DM. Three reference single-dish telescopes are used throughout, calibrated such that ASKAP would just detect a median FRB to **DM = 1000** in the bare Euclidean limit and then made four times fainter in the population. The six sweeps demonstrate how the survey moves between sensitivity-limited and volume-limited regimes, how SFR weighting truncates the high-DM tail of volume-limited surveys, how DM smearing disproportionately affects intermediate-aperture telescopes, and how a steep negative spectral index re-introduces sensitivity limits at every diameter.

1. Euclidean standard candle

We begin in the simplest possible limit: a uniform-density population of identical FRBs in flat, non-expanding space. With intrinsic width $w = 1$ ms and spectral luminosity L_* , the observed flux density is $S = L_*/(4\pi d^2)$, so a telescope with detection threshold $S_{\min}$ has a hard horizon

$$d_{\max} = \sqrt{L_*/4\pi S_{\min}} \propto S_{\min}^{-1/2} \propto D,$$

where the radiometer equation $S_{\min} \propto D^{-2}$ gives the diameter scaling. The expected number of detections per unit distance grows as $dN/dr \propto r^2$ from the uniform density of the population, so the detected DM distribution (with **DM** $\propto d$) is

$$\frac{dN}{dDM} \propto DM^2 \Theta(DM_{\max} - DM), \quad DM_{\max} \propto D.$$

Calibrating the ASKAP horizon to $\text{DM}_{\text{max}}^{\text{ASKAP}} = 1000 \text{ pc cm}^{-3}$ gives Parkes a horizon at $\text{DM}_{\text{max}}^{\text{PKS}} = 5000$ and FAST at $\text{DM}_{\text{max}}^{\text{FAST}} = 25,000$. The integrated event rate scales as $R \propto \Omega_{\text{FoV}} d_{\text{max}}^3 \propto D^{-2} \cdot D^3 = D$: linear in diameter, despite the D^4 collecting-area gain on volume.

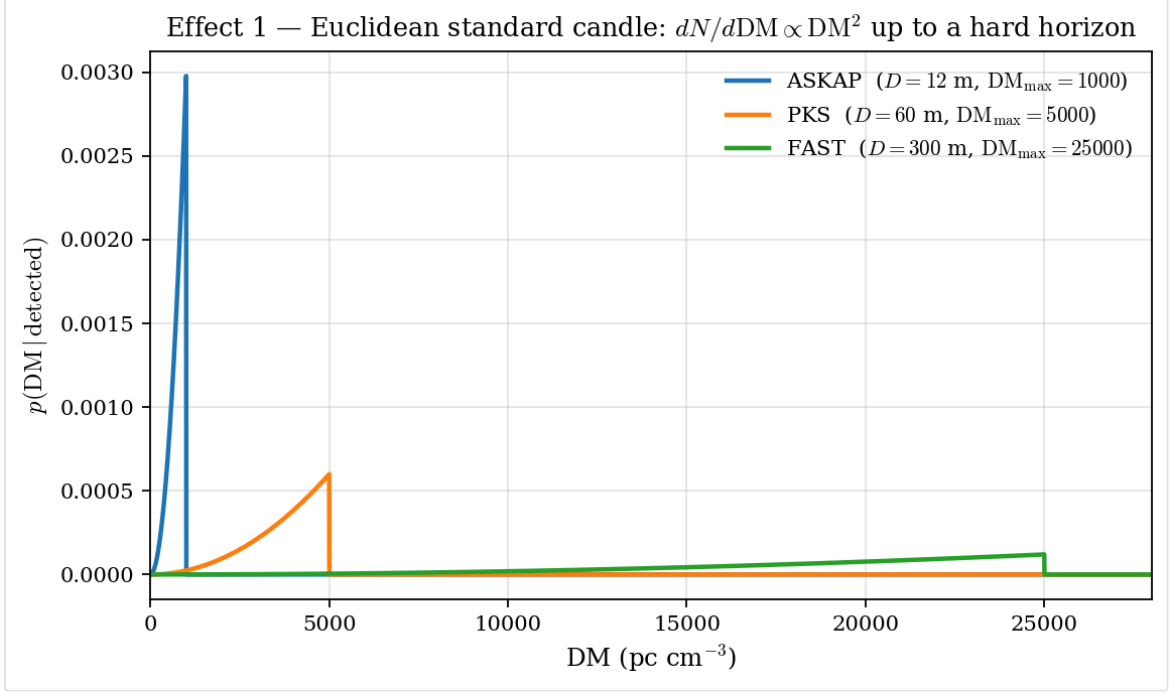


Figure 1. Detected-DM PDF for a Euclidean standard-candle population. Each curve is sharply truncated at $\text{DM}_{\text{max}} \propto D$. There is no high-DM tail whatsoever — every FRB beyond the horizon is invisible.

Takeaway. The Euclidean standard-candle limit gives a hard DM^2 rise to a discrete horizon $\propto D$. Real surveys never look like this; the deviations are precisely what the next four sections quantify.

2. Luminosity function

FRB luminosities are not standard candles. We adopt a log-normal LF in the natural log of the spectral luminosity:

$$\phi(L) dL = \frac{1}{L \sigma \sqrt{2\pi}} \exp \left[-\frac{(\ln L - \ln L_*)^2}{2\sigma^2} \right] dL,$$

with median L_* and natural-log width σ . The detection probability of a source at distance d is the upper tail of the LF above the limiting luminosity at that distance,

$$p_{\text{det}}(d) = \frac{1}{2} \operatorname{erfc} \left[\frac{\ln L_{\min}(d) - \ln L_{\star}}{\sigma\sqrt{2}} \right], \quad L_{\min}(d) = 4\pi d^2 S_{\min}.$$

The hard horizon is replaced by a smooth roll-off whose extent is controlled by σ . In particular, a wider LF (larger σ) lets intrinsically bright bursts be detected from well beyond the standard-candle horizon, and the high-DM tail of a sensitivity-limited survey directly probes σ .

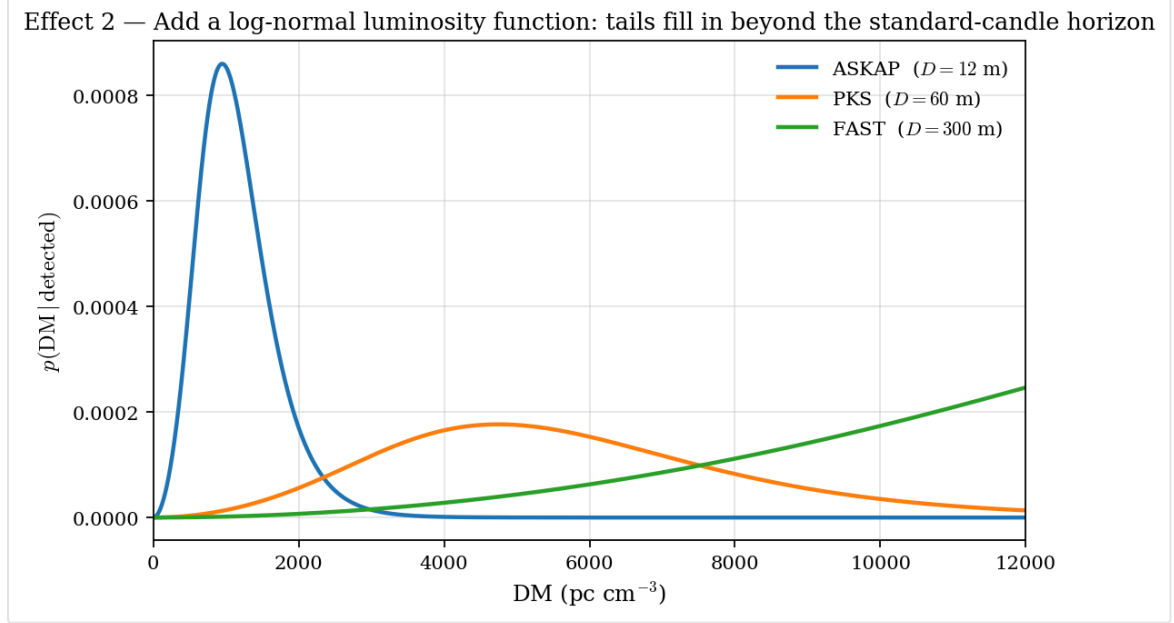


Figure 2. As Figure 1 but with a log-normal LF of width $\sigma_{\ln L} = 1/\sqrt{2}$ added (still Euclidean, no cosmology). Each cutoff softens into a high-DM tail; FAST's horizon — previously at **DM = 25,000** — is now an unphysically broad smear, which is the cue that we must turn on cosmology before this can mean anything.

3. Cosmology

Two changes follow from doing the geometry properly. First, the volume element is no longer Euclidean,

$$\frac{dV_c}{dz} = \frac{4\pi c D_C^2(z)}{H_0 E(z)}, \quad E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda},$$

with comoving distance $D_C(z) = (c/H_0) \int_0^z dz'/E(z')$ and luminosity distance $D_L(z) = (1+z)D_C(z)$. Second, the spectral flux of a source at redshift z acquires a $(1+z)$ factor: $S_\nu = (1+z)L_\nu/(4\pi D_L^2)$. With a flat spectrum (deferred to section 5), the limiting luminosity is

$$L_{\min}(z) = 4\pi D_C^2(z) (1+z) S_{\min}.$$

For the DM--redshift relation we adopt the linear simplification of the Macquart relation, $\mathbf{DM} = 1000 z$. The detected-rate density per unit observer time per unit DM is then the master expression

$$\frac{d\dot{N}}{d\mathbf{DM}} = \frac{\Omega_{\mathbf{FoV}}}{4\pi} \frac{\Phi(z)}{1+z} \frac{dV_c}{dz} p_{\text{det}}(z) \left. \frac{dz}{d\mathbf{DM}} \right|_{z=\mathbf{DM}/1000},$$

where $\Phi(z)$ is the comoving FRB rate density (set to a constant Φ_0 in this section) and the $1/(1+z)$ factor is the cosmological time dilation of the rate. We henceforth plot the area-normalised conditional PDF $p(\mathbf{DM} | \text{detected})$, since $\Omega_{\mathbf{FoV}}$ is independent of DM and cancels in the shape.

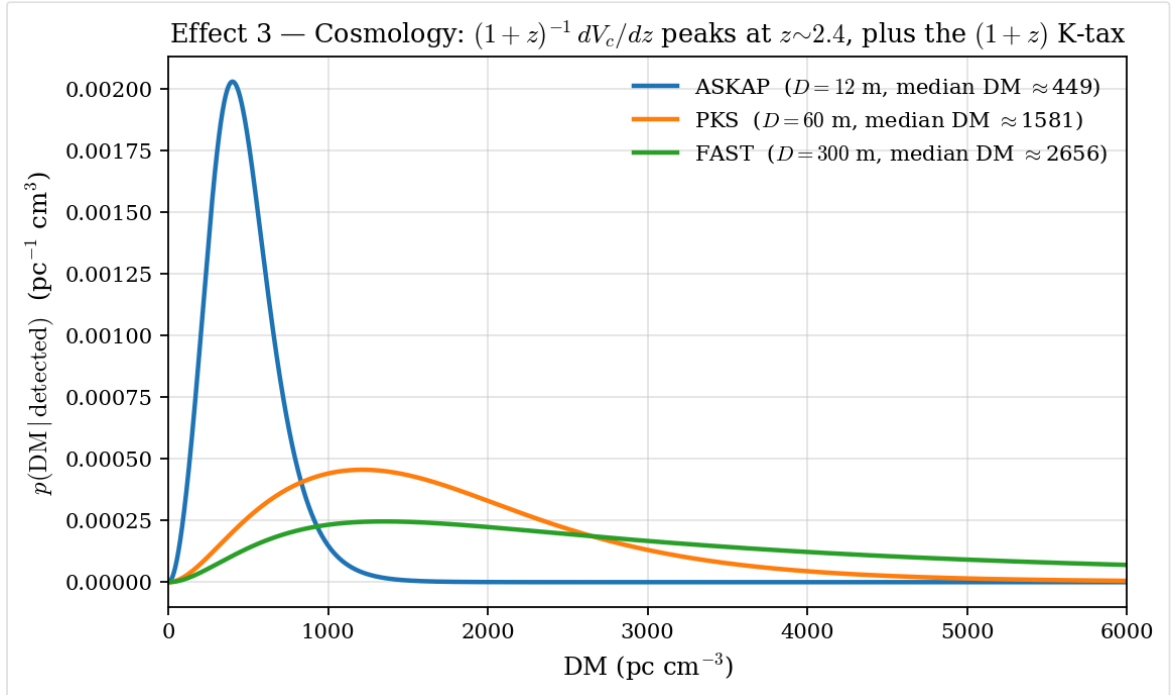


Figure 3. Same population now placed in flat Λ CDM. Cosmology bends the $\mathbf{DM}^2$ rise back over: the volume kernel $(1+z)^{-1} dV_c/dz$ peaks at $z \approx 2.4$ and the $(1+z)$ K-tax slows the rise of $L_{\min}(z)$. ASKAP remains sensitivity-limited; FAST is essentially volume-limited. PKS sits at the cross-over.

Table 1. DM quartiles (pc cm^{-3}), cosmological model with $\sigma_{\ln L} = 1/\sqrt{2}$ and constant Φ_0 .

Telescope	Q_{25}	Q_{50}	Q_{75}	Q_{90}
ASKAP	323	449	600	765
PKS	1020	1581	2315	3178
FAST	1507	2656	4401	6152

Takeaway. The high-DM tail of a sensitivity-limited survey constrains the LF width σ ; a volume-limited survey is largely insensitive to σ and instead probes the cosmological rate density. The diameter of the telescope therefore selects which property the dataset can constrain.

4. Star-formation history

If FRBs track star formation without delay, the comoving rate density is no longer constant but follows the Madau–Dickinson (2014) parametrisation

$$\Phi(z) = \Phi_0 \frac{\psi(z)}{\psi(0)}, \quad \psi(z) = 0.015 \frac{(1+z)^{2.7}}{1 + [(1+z)/2.9]^{5.6}} M_\odot \text{ yr}^{-1} \text{ Mpc}^{-3}.$$

The weight $\psi(z)/\psi(0)$ rises from 1 at $z = 0$ to ≈ 8.9 at cosmic noon ($z \simeq 1.9$), then declines to ≈ 2.1 by $z = 5$. Inserted into the master expression for $d\dot{N}/d\text{DM}$, this preferentially up-weights $z \sim 1\text{--}3$ and down-weights both very low and very high redshifts.

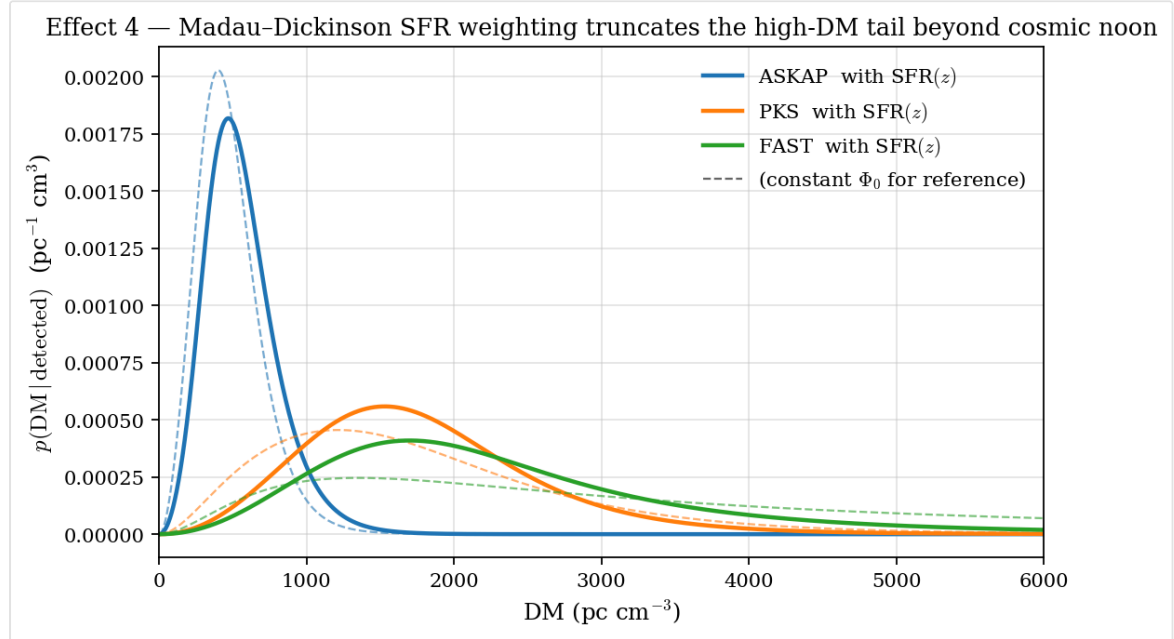


Figure 4. Solid: SFR-weighted distributions. Dashed: constant- Φ_0 baseline from Figure 3. The most striking change is FAST's median dropping from **2656** to **2079** as the high- z tail of the SFR-declined population is amputated past cosmic noon.

Table 2. DM quartiles with versus without SFR weighting.

Telescope	Mode	Q_{25}	Q_{50}	Q_{75}	Q_{90}
ASKAP	const Φ_0	323	449	600	765
ASKAP	SFR(z)	388	529	699	888

Telescope	Mode				
PKS	const Φ_0	1020	1581	2315	3178
PKS	SFR(z)	1228	1687	2228	2827
FAST	const Φ_0	1507	2656	4401	6152
FAST	SFR(z)	1453	2079	2941	4074

Takeaway. SFR weighting chiefly truncates the high-DM tail of *volume-limited* surveys. The contrast between the high-DM tails of PKS- and FAST-class surveys is therefore the cleanest probe of the SFR fall-off beyond cosmic noon.

5. Intra-channel DM smearing

A pulse of dispersion measure $\mathbf{DM}$ traversing a finite frequency channel of width $\Delta\nu_{\text{ch}}$ is broadened in time by an amount proportional to $\mathbf{DM}$. Adding this broadening in quadrature with the intrinsic pulse width gives an effective matched-filter width $w_{\text{eff}} = \sqrt{w^2 + t_{\text{DM}}^2}$, and the radiometer equation gives a detection threshold

$$S_{\text{min}}(\mathbf{DM}) = S_{\text{min}}^{(0)} \sqrt{1 + (\mathbf{DM}/1000)^2},$$

where the normalisation has been chosen so that the smearing time equals the intrinsic 1\,ms width at $\mathbf{DM} = 1000 \text{ pc cm}^{-3}$. The same factor multiplies S_{min} for every telescope — smearing is set by channel width, not dish size — and so the limiting luminosity becomes

$$L_{\text{min}}(z) = 4\pi D_C^2(z) (1+z) S_{\text{min}}^{(0)} \sqrt{1+z^2}.$$

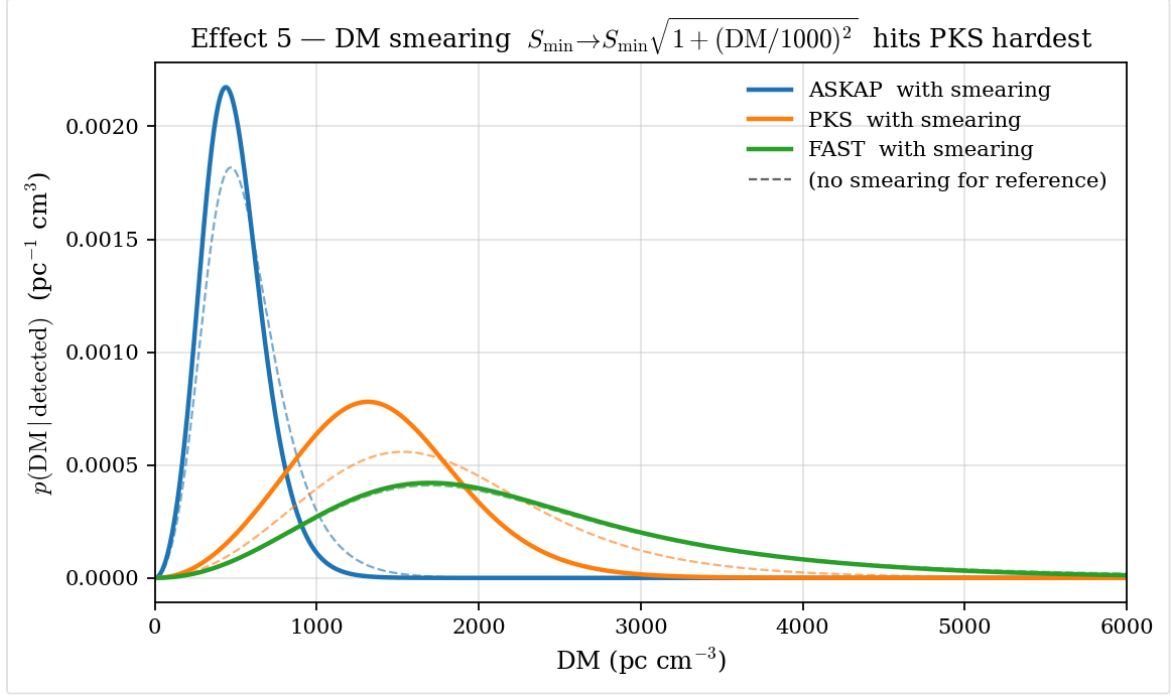


Figure 5. Solid: SFR-weighted distributions with DM smearing applied. Dashed: no smearing, otherwise identical to Figure 4. The largest fractional contraction is for PKS, not FAST — PKS sits exactly in the DM range where the smearing factor is between **1.8** and **3.2**.

Table 3. DM quartiles with versus without intra-channel DM smearing (SFR weighting throughout).

Telescope	Mode	Q_{25}	Q_{50}	Q_{75}	Q_{90}
ASKAP	no smearing	388	529	699	888
ASKAP	with smearing	352	471	605	743
PKS	no smearing	1228	1687	2228	2827
PKS	with smearing	1016	1354	1711	2060
FAST	no smearing	1453	2079	2941	4074
FAST	with smearing	1436	2042	2846	3811

Takeaway. Counter-intuitively, DM smearing affects neither the most nor the least sensitive telescope most strongly — it most strongly affects the intermediate one. ASKAP barely reaches the DM range where smearing matters, and FAST has so much sensitivity headroom that even a $4\times$ threshold degradation at $\mathbf{DM} \approx 4000$ does not push the population below detectability. PKS, however, lives exactly where the smearing factor bites: its Q_{90} contracts by **27%**. Inferences based on the high-DM tail of a Parkes-class survey are therefore highly sensitive to a correct treatment of intra-channel smearing.

6. Spectral index

Real FRBs are not flat-spectrum. With $S_\nu \propto \nu^\alpha$ (so $\alpha < 0$ means fainter at higher frequencies), an FRB observed at ν_o is emitted at $\nu_e = (1+z)\nu_o$ with rest-frame spectral luminosity $L_{\nu_e} = L_{\nu_o} (1+z)^\alpha$. The K-correction modifies the limiting luminosity, now stacking on top of the smearing factor of section 5, to

$$L_{\min}(z) = 4\pi D_C^2(z) (1+z)^{1-\alpha} S_{\min}^{(0)} \sqrt{1+z^2}.$$

For $\alpha = 0$ we recover the previous $(1+z)^1$ scaling. For $\alpha = -2$, the K-correction factor is $(1+z)^3$, giving a **9** $\times$ harder detection threshold at $z = 2$ and a **25** $\times$ harder threshold at $z = 4$. This effectively re-introduces sensitivity-limited behaviour even at FAST.

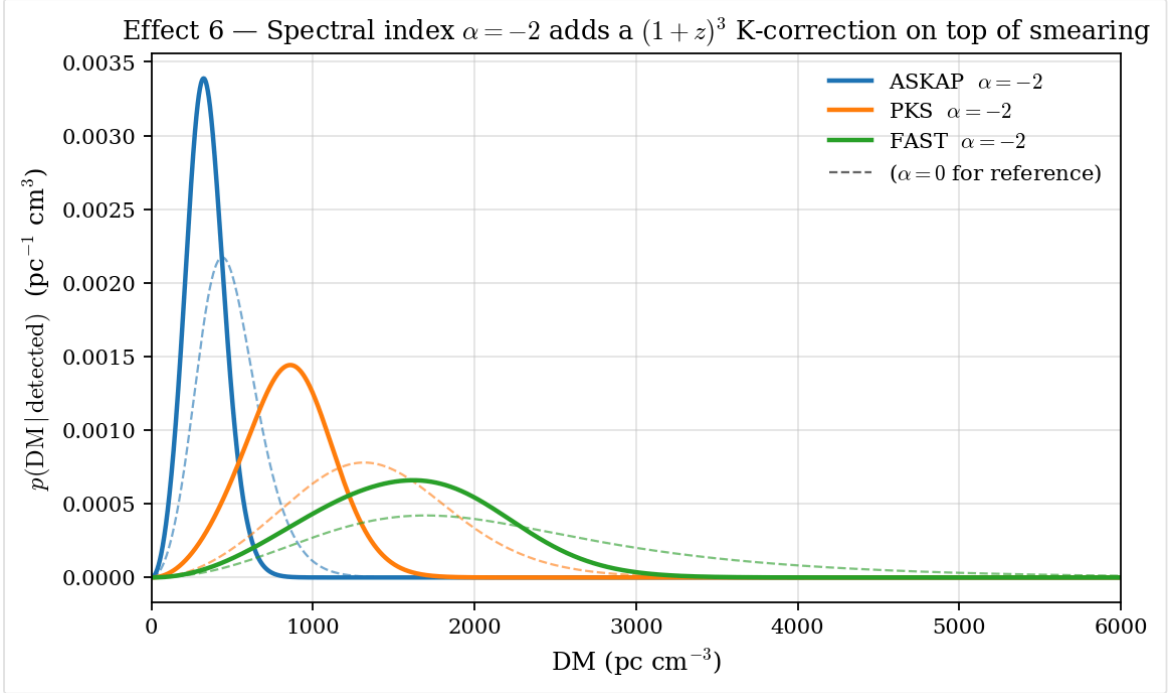


Figure 6. Solid: $\alpha = -2$ on top of SFR weighting and DM smearing. Dashed: $\alpha = 0$ for reference (i.e. the solid curves of Figure 5). Every distribution contracts inwards; FAST's PDF becomes a clean bell shape peaking near **DM** $\sim$ **1600** rather than the broad SFR-shaped distribution of Figure 4.

Table 4. DM quartiles versus the K-correction (SFR weighting and DM smearing applied throughout).

Telescope	α	Q_{25}	Q_{50}	Q_{75}	Q_{90}
ASKAP	0	352	471	605	743
ASKAP	-2	254	332	414	494
PKS	0	1016	1354	1711	2060
PKS	-2	656	847	1032	1199
FAST	0	1436	2042	2846	3811
FAST	-2	1184	1591	1991	2339

Takeaway. A steep negative spectral index acts as a multiplicative “sensitivity tax” that grows with z , capable of converting a volume-limited survey into a sensitivity-limited one. Counterintuitively, this *strengthens* the diagnostic value of comparing small and large dishes: the median-DM ratio FAST/ASKAP rises from **4.3** at $\alpha = 0$ to **4.8** at $\alpha = -2$.

7. Discussion

The six effects act on the detected DM distribution in distinct and partially complementary ways:

- **Euclidean geometry** sets the basic $\mathbf{DM}^2$ rise (uniform-density volume) and the $D \propto \mathbf{DM}_{\max}$ horizon scaling that survives, in modified form, throughout. The hard-horizon limit is unphysical but is the right starting point for intuition.
- **The luminosity function** replaces the hard horizon with a smooth tail whose extent is set by σ . The shape of this tail at high DM is the cleanest observable for the LF width, *provided the survey is sensitivity-limited*; a volume-limited survey is essentially blind to σ .
- **Cosmology** bends the $\mathbf{DM}^2$ rise back over via the $(1+z)^{-1} dV_c/dz$ kernel and slows the rise of $L_{\min}(z)$ by a $(1+z)$ K-tax. The diameter of the telescope determines whether the detected distribution lives below the volume peak (sensitivity-limited) or near/above it (volume-limited).
- **Star-formation history** chiefly truncates the high-DM tail of volume-limited surveys, narrowing FAST's distribution considerably while leaving ASKAP's almost untouched. The PKS–FAST contrast in the high-DM tail is therefore a sensitive probe of SFR(z) beyond cosmic noon.
- **Intra-channel DM smearing** is an instrumental, not astrophysical, effect, but it has the largest impact on intermediate (PKS-class) telescopes — not on the most or least sensitive of the three. It must be included in any forward model that uses the high-DM tail of a Parkes-class survey for inference.
- **Spectral index** re-introduces sensitivity limits at every D via the K-correction $(1+z)^{1-\alpha}$, contracting all distributions inwards and *amplifying*, not suppressing, the diagnostic contrast between small and large dishes.

The model is deliberately minimal. The largest remaining omissions are (i) intrinsic Macquart-relation scatter from host- and IGM-DM contributions; (ii) scatter-broadening from inhomogeneous plasma ($\propto \mathbf{DM}^2$ to $\mathbf{DM}^4$ depending on screen geometry); (iii) a delay-time distribution for FRB progenitors not strictly tracking star formation; (iv) effective receiver multiplicities (PAFs on ASKAP, multibeam on FAST) that change $\Omega_{\mathbf{FoV}}$ for real surveys; and (v) a distribution over α rather than a single value. None of these change the qualitative conclusions above; all of them would smear the curves further.

Generated by the Python scripts `frb_doc_figures.py`, `frb_dndm_shapes.py`,
`frb_dndm_sfr.py`, `frb_dndm_smear.py`, and `frb_dndm_alpha.py` in `/Users/mbailes`.
Cosmology assumed: flat Λ CDM with $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.315$.